

Course: Information Assurance and Security

Lab Title: Assignment#2

Group Members: Joshua Il B Dajao, Vonzelle Cuizon

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Instructor: Engr. Godwin Monserate

1. What is Digital Forensics?

Digital Forensics is the scientific process of identifying, preserving, analyzing, and presenting digital evidence from electronic devices in a legally admissible manner. It involves investigating data breaches, cybercrimes, and unauthorized access to ensure accountability and support legal proceedings.

2. Why is There a Need for Digital Forensic Services?

- **Cybercrime Investigation:** To uncover digital evidence in cases of hacking, fraud, or data theft.
- **Legal Compliance:** Ensures evidence meets court standards.
- **Incident Response:** Helps organizations respond to security breaches.
- **Data Recovery:** Retrieves lost or deleted files.
- **Corporate Investigations:** Detects insider threats or policy violations.

3. Where is ESI (Electronically Stored Information) Commonly Stored?

ESI is stored in:

- Hard drives, SSDs, USB drives
- Cloud storage (Google Drive, Dropbox)
- Mobile devices (phones, tablets)
- Emails and messaging apps
- Network logs and servers
- IoT devices (smart cameras, wearables)

4. Circumstances Where Digital Forensics is Applied

- **Cyberattacks** (ransomware, phishing)
- **Employee misconduct** (data leaks, insider threats)
- **Legal disputes** (intellectual property theft)
- **Fraud investigations** (financial crimes)
- **Law enforcement** (child exploitation, terrorism)

5. Training Required to Become a Digital Forensic Examiner

- **Certifications:** CISSP, CHFI, EnCE, CFCE
- **Technical Skills:** File system analysis, malware reverse engineering
- **Legal Knowledge:** Chain of custody, evidence handling
- **Hands-on Experience:** Using tools like FTK, EnCase, Autopsy

6. What Does Digital Forensic Examination Provide?

- **Evidence identification** (deleted files, logs)
- **Timeline reconstruction** (user activity tracking)
- **Malware detection** (rootkits, spyware)
- **Legal reports** (court-admissible findings)

7. Difference Between Digital Forensics and Data Recovery

Digital Forensics	Data Recovery
Focuses on legal evidence	Recovers lost data without legal concerns
Maintains chain of custody	Does not prioritize evidence integrity
Used in criminal cases	Used for accidental deletions

8. What to Do If Evidence Exists on a Computer?

- Do NOT alter the system (shut it down if necessary).
- Document everything (photograph, notes).
- Contact a forensic expert to preserve evidence properly.

9. Types of Legal Cases in Digital Forensics

- Criminal Cases (hacking, fraud)
- Civil Litigation (IP theft, employment disputes)
- Corporate Investigations (insider threats)
- Cybersecurity Incidents (data breaches)

10. Definitions of Key Certifications

- **CISSP** (Certified Information Systems Security Professional): Advanced cybersecurity certification.
- **MCSE** (Microsoft Certified Solutions Expert): Microsoft-based IT infrastructure expertise.

- **CEH** (Certified Ethical Hacker): Penetration testing and vulnerability assessment.
- **CFCE** (Certified Forensic Computer Examiner): Digital forensic investigation.
- **CEECS** (Certified Electronic Evidence Collection Specialist): Proper evidence collection.
- **CHFI** (Computer Hacking Forensic Investigator): Cybercrime investigation.
- **CCFE** (Certified Computer Forensics Examiner): Forensic analysis techniques.
- **EnCE** (EnCase Certified Examiner): Proficient in EnCase forensic tools.

11. Risks of Not Consulting a Forensic Expert Early

- **Evidence tampering** (accidental modifications).
- **Data loss** (overwritten files).
- **Legal invalidation** (improper handling).

12. Should In-House Experts Handle Evidence First?

No. Untrained personnel may compromise evidence. Always involve a forensic expert immediately.

13. What Makes a Collection Procedure Forensically Sound?

- Write-blocking tools (prevents data alteration).
- Chain of custody documentation.
- Hashing (ensures data integrity).

14. Proper Procedures If a Computer Contains Evidence

1. Isolate the device (disconnect from networks).
2. Avoid using the system (prevents overwriting).
3. Call a forensic investigator.

15. Process of a Digital Forensic Examination

1. **Identification** (locate evidence sources).
2. **Preservation** (secure and document).
3. **Analysis** (extract and interpret data).
4. **Documentation** (detailed report).
5. **Presentation** (court testimony if needed).

16. When to Use Computer Forensics?

- Suspected data breaches.
- Employee misconduct.
- Legal disputes requiring digital evidence.

17. Can Anyone Recover Lost Data?

Only trained professionals should attempt recovery to avoid permanent loss. But if its for personal reasons, one can easily learn from education/tutorial websites like reddit or youtube.

18. Can Data Be Recovered from SMS Apps?

Yes, if not overwritten. Forensic tools can retrieve deleted messages.

19. What to Do If You Were Hacked?

1. Disconnect the system.
2. Preserve logs.
3. Report to authorities.
4. Engage a forensic investigator.

20. How to Create a Secure Password?

- **Use 12+ characters** (mix letters, numbers, symbols).
- **Avoid common words.**
- **Enable MFA** (Multi-Factor Authentication).

21. Why Should I Care About Security If I'm Not a Big Target?

Yes we all should. Small targets are often easier to hack. Attackers exploit weak security for larger attacks.

22. Can Data Be Recovered from a Damaged Hard Drive?

Sometimes, if platters are intact. Professional recovery services are needed.

23. Can Blockchain Be Hacked?

Extremely difficult, but exchanges/wallets can be compromised.

24. How to Prevent Being Hacked?

- Update software regularly.
- Use strong passwords + MFA.
- Avoid phishing scams.
- Regular backups.

25. Should I Insert a Found USB into My Computer?

Never! It could contain malware. Hand it over to IT or security personnel.